

# Payroll Output Converter

## Technical Overview & Migration Handover

Prepared by Jamie Callawick · October 2026

### 1. The Problem We Are Solving

New Simplifi clients need their staff loaded before go-live. Today that means one of two things:

- The client exports an employee report from their payroll system (for example IMS/MYOB) and sends it to the delivery team, who reshape it by hand into the Simplifi Staff Import Template.
- Or the client is asked to fill in the template themselves — renaming columns, splitting names and addresses, removing leavers, converting dates — in Excel.

Both routes are slow and error-prone. Payroll exports contain far more than Simplifi needs (IRD numbers, bank accounts, KiwiSaver settings), use different column names, combine fields Simplifi keeps separate (e.g. "HAWERA, 4610"), and include staff who have already left. Every manual step risks a wrong mapping, a missed leaver, or sensitive data being emailed around unnecessarily.

*The Payroll Output Converter is the customer-facing version of the delivery team's internal Customer Data Cleanser. It lets the client produce a clean, validated Simplifi import file themselves, in their own browser, without sending their payroll export to anyone.*

### 2. What the Tool Does

The Payroll Output Converter is a lightweight, single-page web application. A payroll administrator:

- Selects their payroll provider (IMS / MYOB, or "Other payroll system").
- Drops or browses to their employee export — CSV, XLSX or XLS.
- Reviews the column mapping. Known formats (IMS) are mapped automatically; other files are matched by column name and the rest can be mapped from a dropdown.
- Chooses options: include leave balances, exclude leavers, pay rate valid-from date, name and address handling.

- Reviews a validated preview (errors and warnings highlighted per cell) and downloads a Simplifi-ready CSV in one click.
- Optionally saves the setup (mapping, leave codes, options) so the next conversion is a single click.

The layout, branding and navigation match the Leave Balance Converter; a nav bar at the top links the two tools.

### 3. Technical Architecture

The application follows the same deliberately simple design as the Leave Balance Converter: one static file, no backend and no build step.

|                        |                                                                                                                                                                                                     |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Type</b>            | Static single-page application (SPA)                                                                                                                                                                |
| <b>Language</b>        | Vanilla HTML5, CSS3 and JavaScript — no framework, no build step                                                                                                                                    |
| <b>File count</b>      | One file: index.html (all logic, styles, markup and the Excel reader are self-contained)                                                                                                            |
| <b>Dependencies</b>    | One: SheetJS (xlsx) 0.18.5, embedded inline in index.html to read XLSX/XLS files. It is not loaded from a CDN, so the page makes no external requests. CSV files are parsed by the tool's own code. |
| <b>Data processing</b> | All parsing, mapping, transformation and validation runs in the user's browser (FileReader, TextDecoder, Blob, URL.createObjectURL)                                                                 |
| <b>Storage</b>         | Optional: saved setups in the browser's localStorage (column mapping and settings only — no employee data). See section 5.                                                                          |
| <b>Server</b>          | None — no backend, no API calls, no server-side logic. (The internal cleanser's AI mapping call was removed for this version.)                                                                      |
| <b>Browser support</b> | All modern browsers (Chrome, Edge, Firefox, Safari)                                                                                                                                                 |

Processing is a five-step wizard: Upload → Map Columns → Options → Preview → Download. The main building blocks in the JavaScript are:

- **TARGET\_HEADERS** — the 42 columns of the Simplifi Staff Import Template, in template order. Output columns always follow this order.
- **PRESETS** — known payroll formats. Each preset has a header signature (detected when at least 75% of the signature columns are present) and a fixed column map. Currently: IMS Payroll employee export. New providers are added as new PRESETS entries.
- **SYNONYMS** — built-in header matching for files with no preset (e.g. "Surname" → Last Name, "Staff ID" → Employee Code). Never maps two columns to the same field.

- **SPECIAL\_TARGETS** — mappings that produce several output columns from one source: split full name, split street, split "TOWN, 1234", finish date → Last Date Employment + Is Active, and leave balance → one row per leave type.
- **VALIDATION\_RULES** — field rules taken from the template's Header Descriptions sheet (required, max length, email, boolean, date, number), plus cross-field checks.

## 4. Conversion Logic

### 4.1 Reading the file

- XLSX/XLS are read with SheetJS (dates formatted dd/mm/yyyy). CSV is read as UTF-8, falling back to Windows-1252 — IMS CSVs are Windows-1252 and contain non-breaking spaces in some headings, which are normalised to spaces.
- The CSV parser handles quoted values, including commas and line breaks inside quotes.
- The headings row is auto-detected within the first 10 rows (scored on label-like text and payroll keywords). Title rows above it — such as the IMS "EMPLOYEE / STANDARD / LEAVE" group row — are skipped. The user can override the row.
- IMS layout handling: IMS puts a "+" marker in column A with "Employee ID" heading over it. If the heading sits over merged cells, the column with the most distinct values is used. If the headings are shifted one column left of the data (column A holds only markers), all headings are shifted right by one.

### 4.2 IMS / MYOB preset mapping

| IMS column                   | Simplifi field(s)           | Notes                                                                         |
|------------------------------|-----------------------------|-------------------------------------------------------------------------------|
| <b>Emp No</b>                | Employee Code               | Payroll number, not the internal Employee ID                                  |
| <b>First Names / Surname</b> | First Name / Last Name      |                                                                               |
| <b>E-mail</b>                | Email Address               |                                                                               |
| <b>Mobile #</b>              | Phone                       | Daytime # is not used                                                         |
| <b>Address 1</b>             | Street Number + Street Name | "3/212 High Street" → "3/212" + "High Street"                                 |
| <b>Address 2, Address 3</b>  | Suburb                      | First non-blank wins; a "TOWN, 1234" value is routed to Town/City + Post Code |
| <b>Address 4</b>             | Town/City + Post Code       | "HAWERA, 4610" → "Hawera" + "4610" (upper case is title-cased)                |

| IMS column                                  | Simplifi field(s)                                          | Notes                                             |
|---------------------------------------------|------------------------------------------------------------|---------------------------------------------------|
| <b>Occupation</b>                           | Job Role(s)                                                |                                                   |
| <b>Pay Point</b>                            | Department                                                 |                                                   |
| <b>Costing Code</b>                         | Employee Cost Code                                         |                                                   |
| <b>Finish Date</b>                          | Pay Roll - Last Date<br>Employment + Is Active<br>Employee | Is Active = FALSE when the date is<br>in the past |
| <b>Pay Frequency</b>                        | Pay Roll - Calendar                                        |                                                   |
| <b>Hourly Rate</b>                          | Pay Roll - Pay Rate                                        | Valid-from date added (see 4.4)                   |
| <b>AL / SL / AH / TIL /<br/>FVL Balance</b> | Leave - Balance Owing + Leave<br>- Type Code               | One row per leave type (see 4.3)                  |

Deliberately skipped: Employee ID (internal IMS ID), Full Name, Preferred Name, Daytime #, IRD No, tax, bank account, birth date, age, gender, KiwiSaver, child support and leave entitlement settings. Users can still map any skipped column manually.

### 4.3 Leave balances (toggle, on by default)

The template holds one leave pair (Leave - Balance Owing, Leave - Type Code) per row, and additional values go on separate lines. The converter therefore writes one row per employee per leave type, repeating the employee's full details on every row. Employees with no leave balances get a single row with the leave columns blank.

| IMS field                                                | Default code  | Default                              |
|----------------------------------------------------------|---------------|--------------------------------------|
| <b>AL Balance</b>                                        | AL            | On                                   |
| <b>SL Balance</b>                                        | SL            | On                                   |
| <b>AH Balance</b>                                        | AH            | On                                   |
| <b>TIL Balance</b>                                       | TIL           | On                                   |
| <b>FVL Balance</b>                                       | FVL           | On                                   |
| <b>AL Outstanding, AL Accrued, AL in Advance</b>         | AL            | Off — already included in AL Balance |
| <b>PL Outstanding / CL Outstanding / SVL Outstanding</b> | PL / CL / SVL | Off                                  |

- Codes are editable in the Options step and must match the leave type codes set up for the client's payroll in Simplifi.

- Balances are exported exactly as in the file (currency symbols and thousands separators stripped) — no unit conversion. The units column shows the IMS "<type> Units In" value; mixed units (e.g. some staff in hours, some in days) raise a warning.
- "Skip zero or empty balances" (on by default) drops zero-balance rows. Blank balances are always skipped.
- A warning is shown if two included balances share a code; an error if a code is blank.
- For other providers, any column can be mapped to "Leave balance"; headers such as "Annual Leave Balance" are suggested automatically.

#### 4.4 Guardrails

|                                |                                                                                                                                                                                                                                                                                    |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exclude leavers</b>         | On by default. Removes staff whose Last Date Employment is today or earlier. For IMS files, rows without the "+" active marker in column A are also treated as leavers (same rule as the Leave Balance Converter). The number excluded is shown on the Preview and Download steps. |
| <b>Pay rate valid-from</b>     | The template requires a valid-from date whenever a pay rate is imported. The tool fills it (default: today, editable) where the file doesn't supply one, and flags any pay rate without a date as an error.                                                                        |
| <b>Duplicate Employee Code</b> | The same Employee Code on two different staff members is an error. (Repeated codes on one person's leave rows are expected.)                                                                                                                                                       |
| <b>Leave rows</b>              | A leave balance without a leave code, or without an Employee Code, is an error.                                                                                                                                                                                                    |
| <b>Field rules</b>             | Required First/Last Name; max lengths; valid email; TRUE/FALSE booleans; dd/MM/yyyy dates; numeric post code, balances and hours — all from the template's Header Descriptions.                                                                                                    |
| <b>Export with issues</b>      | Rows with errors are still exported but clearly flagged, with a filter to show only rows with issues.                                                                                                                                                                              |

## 5. Saved Setups

Users can save their choices so the next conversion needs no re-mapping. A setup contains only:

- The column → Simplifi field mapping, keyed by column name, plus the list of column names in the file.
- Leave balance selections and leave type codes.
- Option toggles, and the pay rate date mode ("today" is stored as "use the day of conversion"; a chosen date is stored as fixed).

|                             |                                                                                                                                                                                                                                                                                               |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Save in this browser</b> | Stored in localStorage under simplifi.payrollOutputConverter.setups (namespaced, because the Leave Balance Converter shares the same github.io origin). Up to 25 setups. When a file whose columns match a saved setup by 80% or more reaches Map Columns, the user is offered "Apply setup". |
| <b>Download setup file</b>  | Downloads <name>.simplifi-setup.json. "Load setup file" on the upload step applies it on any computer. The delivery team can prepare a setup and send it to a client before go-live.                                                                                                          |
| <b>Safety</b>               | Loaded files are validated: wrong app, unknown fields and invalid targets are discarded. Columns in the setup but missing from the file are listed; new columns are marked for review.                                                                                                        |
| <b>Limits</b>               | Browser-saved setups are lost if the user clears browser data, uses private browsing, or changes computer/browser. The downloaded file is the durable copy.                                                                                                                                   |

## 6. Security & Data Privacy

*Unlike the Leave Balance Converter, this tool does handle personal information — the output contains names, contact details, addresses, pay rates and leave balances. All of it is processed only in the user's browser. Nothing from the uploaded file is transmitted or stored.*

|                         |                                                                                                                                                                                                        |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Network requests</b> | Zero after page load. No analytics, telemetry, fonts or third-party scripts. The Excel reader is embedded in the page.                                                                                 |
| <b>PII handling</b>     | File contents are read by the FileReader API into JavaScript memory only. Sensitive payroll fields (IRD number, bank account, birth date, gender, KiwiSaver) are skipped by the IMS preset by default. |
| <b>Data retention</b>   | None for employee data — discarded when the tab closes or the user clicks "Start over". Saved setups contain column names, codes and settings only.                                                    |
| <b>Authentication</b>   | Not required. Access control, if needed, is handled at the hosting layer.                                                                                                                              |
| <b>File upload</b>      | No upload occurs. The "upload" is a local file read.                                                                                                                                                   |
| <b>Output file</b>      | The downloaded CSV contains the client's staff PII and should be handled under the client's and Simplifi's normal data-handling policies.                                                              |

|              |                                                                                                                                                                                                                                                                                               |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dependencies | SheetJS 0.18.5 (embedded). This version has published advisories (prototype pollution, CVE-2023-30533, fixed in 0.19.3; ReDoS, CVE-2024-22363, fixed in 0.20.2). Exposure is limited because files are the user's own and nothing runs server-side, but upgrading is recommended (section 9). |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

The footer of the application displays: "All data is processed locally in your browser — nothing is sent to any server."

## 7. Current Deployment

|                    |                                                                                                                                                                                                                |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Repository         | <a href="https://github.com/Jamie-Callawick/Payroll-Output-Converter">https://github.com/Jamie-Callawick/Payroll-Output-Converter</a>                                                                          |
| Owner              | Personal account — Jamie Callawick (Jamie-Callawick)                                                                                                                                                           |
| Branches           | main (production), dev (development / staging)                                                                                                                                                                 |
| Hosting            | GitHub Pages — served as a static site directly from the repository                                                                                                                                            |
| URL format         | <a href="https://jamie-callawick.github.io/Payroll-Output-Converter/">https://jamie-callawick.github.io/Payroll-Output-Converter/</a> (planned)                                                                |
| Linked tool        | Leave Balance Converter — <a href="https://jamie-callawick.github.io/Leave-Balance-Worker-App/">https://jamie-callawick.github.io/Leave-Balance-Worker-App/</a> (both tools link to each other in the top nav) |
| Excluded from repo | Reference Files/ (sample payroll exports and the Simplifi template) is listed in .gitignore and must never be committed — it can contain customer data.                                                        |
| Build process      | None — changes pushed to main are immediately live. No CI/CD pipeline is configured.                                                                                                                           |

## 8. Migration: Personal GitHub to Simplifi Organisation

Same process as the Leave Balance Converter; ideally both repositories move together so their nav links can be updated at the same time. No code changes are required apart from the nav URLs.

|   |                       |                                                                                                                             |
|---|-----------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 1 | Create org repository | Create Payroll-Output-Converter (or equivalent) in the Simplifi GitHub organisation. Private recommended during transition. |
|---|-----------------------|-----------------------------------------------------------------------------------------------------------------------------|

|          |                                 |                                                                                                                                                                                                                      |
|----------|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>2</b> | <b>Transfer or mirror</b>       | Option A (recommended): Settings > Danger Zone > Transfer ownership to the Simplifi org — keeps history and branches. Option B: clone and push to a new org repo. Confirm Reference Files/ has never been committed. |
| <b>3</b> | <b>Update GitHub Pages</b>      | Settings > Pages > Source: main branch (root). The URL becomes https://<org>.github.io/Payroll-Output-Converter/.                                                                                                    |
| <b>4</b> | <b>Update nav links</b>         | Update the Leave Balance Converter link in this tool's top nav, and this tool's link in the Leave Balance Converter, to the new URLs.                                                                                |
| <b>5</b> | <b>Saved setups notice</b>      | Browser-saved setups are tied to the site address, so they will not carry over to a new URL. Let clients know to download their setup file before cutover, or re-save after.                                         |
| <b>6</b> | <b>Branch protection</b>        | Protect main: require pull request review, disable force-push.                                                                                                                                                       |
| <b>7</b> | <b>Team access</b>              | Add Simplifi dev team members (Write or Maintain).                                                                                                                                                                   |
| <b>8</b> | <b>Custom domain (optional)</b> | e.g. tools.simplifi.work/payroll-converter — add a CNAME file and DNS record per GitHub Pages docs.                                                                                                                  |
| <b>9</b> | <b>Verify &amp; smoke test</b>  | Load the new URL, convert a sample IMS XLS and CSV, check leave rows and leaver exclusion, save and re-apply a setup, and check the browser console for errors.                                                      |

## 9. Risks & Mitigations

| <b>Risk</b>                            | <b>Likelihood</b> | <b>Impact</b> | <b>Mitigation</b>                                                                                                                                                                              |
|----------------------------------------|-------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Outdated SheetJS version</b>        | Medium            | Medium        | Embedded SheetJS 0.18.5 has known advisories. Replace with the current release from cdn.sheetjs.com (the npm package is no longer updated), keeping it embedded, and re-test XLS/XLSX reading. |
| <b>Customer data committed to repo</b> | Low               | High          | Reference Files/ is git-ignored. Keep test exports outside the repo; review pull requests for data files.                                                                                      |

| Risk                                | Likelihood | Impact | Mitigation                                                                                                                                                                                               |
|-------------------------------------|------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Wrong leave codes or units</b>   | Medium     | High   | Codes default to IMS abbreviations but must match the client's Simplifi leave types. Units are shown and mixed units are warned about. Delivery team should confirm codes (a prepared setup file helps). |
| <b>IMS export format changes</b>    | Low        | Medium | Detection needs 75% of signature columns; unmatched columns fall back to name matching with a visible warning. Update the PRESETS entry when IMS changes.                                                |
| <b>Personal account access lost</b> | Low        | High   | Complete migration to the Simplifi org promptly; keep a local backup of index.html.                                                                                                                      |
| <b>URL change post-migration</b>    | High       | Medium | Old GitHub Pages URLs do not redirect, and browser-saved setups stay on the old address. Communicate the new URL and ask users to download setup files before cutover.                                   |
| <b>No automated tests</b>           | Medium     | Medium | Add a GitHub Actions workflow that runs the page headlessly (Playwright) against sample files and compares output on each pull request.                                                                  |
| <b>Publicly accessible URL</b>      | Medium     | Low    | No data is sent anywhere, so exposure is low. Cloudflare Access or org-only Pages can restrict access without code changes.                                                                              |
| <b>No dedicated maintainer</b>      | Medium     | Medium | Assign a named owner for both converters and add the repos to team monitoring.                                                                                                                           |

## 10. Testing

Tested headlessly in Chromium (Playwright) with all network requests blocked, confirming the page works with no external access.

|                                     |                                                                                                                                                                                                                    |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>TRINITY (IMS XLS, 106 staff)</b> | IMS preset detected; 24 leavers excluded; 82 staff → 225 import rows with leave. Staff fields identical to the internal Customer Data Cleanser output for all 82 staff. Mixed AL units detected (4 staff in days). |
| <b>IMS CSV variants</b>             | Windows-1252 CSV with the group heading row, and a variant with headings shifted one column left — both produce the same output as the XLS. Rows without the "+" marker excluded as leavers.                       |

|                     |                                                                                                                                                                                                                                                   |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Generic CSV</b>  | Title rows skipped, headings found on row 3, "SURNAME, Given" split, leave columns suggested, duplicate Employee Code and invalid email flagged.                                                                                                  |
| <b>Saved setups</b> | Saved in browser and as a file; re-applied to the same file gives identical output (250 rows — that test setup kept leavers in); applied on a fresh browser from the file; non-setup JSON rejected; different file layouts not offered the setup. |

## 11. Future Considerations

- More payroll providers — add PRESETS entries (e.g. Smartly, Rocket, Crystal Payroll) once sample employee exports are available, mirroring the Leave Balance Converter.
- Upgrade SheetJS to the current release (see section 9).
- Contracted hours — the template supports contracted hours rows; IMS "Hours Per Period" could feed Contracted Hours - Number Hours once the rules are agreed.
- CI/CD — a GitHub Actions smoke test on pull requests.
- Version number in the footer (e.g. v1.1.0) to help support identify which version a client is running.
- Shared shell — the two converters could become one site with a shared header and a tool switcher once both are hosted together.

## 12. Change Log

Changes made to the converter, newest first. Add a new entry for each release.

|                                |                                                                                                    |
|--------------------------------|----------------------------------------------------------------------------------------------------|
| <b>Date</b>                    | 1 October 2026                                                                                     |
| <b>Change</b>                  | Saved setups: "Save in this browser", "Download setup file" and "Load setup file".                 |
| <b>Reason</b>                  | So clients and the delivery team don't have to re-map columns and re-enter leave codes every time. |
| <b>Conversion logic</b>        | Unchanged. A setup only pre-fills the mapping, leave codes and options.                            |
| <b>Backwards compatibility</b> | No change to output when no setup is used.                                                         |
| <b>Tested with</b>             | TRINITY IMS XLS and CSV — see section 10.                                                          |

|                                      |                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Date</b>                          | 1 October 2026                                                                                                                                                                                                                                                                                                       |
| <b>Change</b>                        | Initial release — customer-facing port of the internal Customer Data Cleanser.                                                                                                                                                                                                                                       |
| <b>Differences from the cleanser</b> | Leave Balance Converter branding and nav; AI mapping call removed; leave balances added (toggle, on by default); 42-column template; pay rate valid-from guardrail; duplicate Employee Code check; IMS "+" marker leaver rule; Windows-1252 CSV support; shifted-heading detection; output in template column order. |
| <b>Tested with</b>                   | TRINITY IMS XLS (106 staff) — staff output identical to the cleanser.                                                                                                                                                                                                                                                |

## 13. Point of Contact

For questions about the codebase, the current deployment, or the migration process, contact:

|                       |                            |
|-----------------------|----------------------------|
| <b>Owner / Author</b> | Jamie Callawick            |
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| <b>GitHub</b>         | github.com/Jamie-Callawick |

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